MATH70076: Data Science - Formative Assessment

MSc in Statistics 2026/27, Imperial College London

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Deadline: Monday 12 October 2026 at 13:00.

For this formative assessment you should submit two files via the Imperial College VLE on Canvas by the deadline stated above. Your files should be named as follows:

- `YOURCID-MATH70076-formative-assessment.pdf`: your rendered report,
- `YOURCID-MATH70076-formative-assessment.zip`: a zip file containing the relevant source code to generate your report.

All submitted materials should be clearly presented and be understandable as stand-alone documents.

Please note that large files can take quite some time to upload. Ensure that you upload each document to the correct part of the learning space in a timely manner.

This mock coursework is a formative assessment and so will not contribute to your overall grade for the module. It is designed for you to practice the skills you have learned in the first workshop and to get feedback on your reflective writing style. This activity is expected to take approximately 6 hours of individual effort. Assessment criteria are given in the "set yourself up for success" boxes. Satisfying 18 or more out of these 23 criteria corresponds to a notional "pass" grade.

In submitting this mock assessment you certify that it is entirely your own work, apart from where otherwise acknowledged, and includes no plagiarism. Note that software tools are used as part of plagiarism detection.

Background

Generalised Pareto Distribution

The Generalized Pareto Distribution (GPD) is a flexible family of continuous probability distributions that arises naturally in extreme value theory, particularly for modelling the distribution of excesses over a threshold. It is parametrised by a shape parameter $\xi \in \mathbb{R}$, a scale parameter $\sigma > 0$, and a location parameter $u \in \mathbb{R}$. Its cumulative distribution function (CDF) is given by

$$F(x;\, \sigma, \xi, u) = \begin{cases} 1 - \left(1 + \dfrac{\xi(x-u)}{\sigma}\right)_+^{-1/\xi}, & \xi \neq 0,\ x \geq u; \\ 1 - \exp\!\Big(-\dfrac{x-u}{\sigma}\Big), & \xi = 0,\ x \geq u; \end{cases} \tag{1}$$

where $x_+ = \max(x, 0)$ and its probability density function (PDF) is

$$f(x; \sigma, \xi, u) = \begin{cases} \dfrac{1}{\sigma}\left(1 + \dfrac{\xi(x-u)}{\sigma}\right)_+^{-1/\xi - 1}, & \xi \neq 0, \\ \dfrac{1}{\sigma}\exp\!\Big(-\dfrac{x-u}{\sigma}\Big), & \xi = 0, \end{cases} \tag{2}$$

defined on the same support as the CDF. The GPD encompasses a variety of tail behaviours:

- when $\xi > 0$ the GPD has heavy, slowly decaying tails,
- in limiting case when $\xi \to 0$ the GPD reduces to an exponential distribution;
- when $\xi < 0$ the GPD has light, quickly decaying tails with a finite upper endpoint of $x^+ = u - \sigma/\xi$.

Probability Integral Transform

The probability integral transform states that if a random variable X has a continuous cumulative distribution function $F_X(x)$, then the transformed variable $A = F_X(X)$ follows a uniform distribution on $[0, 1]$. Conversely, if F is an invertible function then $Y = F_X^{-1}(A)$ has the same distribution as X.

Questions

Question 1

Derive an expression for the inverse cumulative distribution function (also known as the quantile function) $F_X^{-1} : [0,1] \to [u, x^+]$ of $X \sim \text{GPD}(u, \sigma, \xi)$. Your answer should refer to at least one equation given in the background material.

Solution to Question 1

We work out the two separate cases where $\xi = 0$ and where $\xi \neq 0$.

Case 1

When $\xi = 0$, $F(x; \sigma, \xi, u) = 1 - \exp\left(-\frac{x-u}{\sigma}\right)$.

We set

$$\begin{aligned} p &= 1 - \exp\left(-\frac{x-u}{\sigma}\right) \\ \exp\left(-\frac{x-u}{\sigma}\right) &= 1 - p \\ \frac{x-u}{\sigma} &= -\log(1-p) \\ x - u &= -\sigma \log(1-p) \end{aligned}$$

and then we finally obtain $F_X^{-1}(p) = u - \sigma \log(1-p)$.

When we check the correctness by setting $p = 0$ and $p = 1$ respectively in the quantile function obtained above, we obtain $x = u$ and $x = +\infty$,which are the correct endpoints.

Case 2

When $\xi \neq 0$, $F(x; \sigma, \xi, u) = 1 - \left(1 + \frac{\xi(x-u)}{\sigma}\right)_+^{-1/\xi}$.

We set the above CDF to p, and then obtain the quantile function $F_X^{-1}(p) = \frac{\sigma}{\xi}(1-p)^{-\xi} - \frac{\sigma}{\xi} + u$.

At $p = 0$ and $p = 1$, $F_X^{-1}(p)$ returns u and x^+ respectively.

Putting it altogether, the inverse cumulative distribution function is

$$F_X^{-1}(p) = \begin{cases} \dfrac{\sigma}{\xi}(1-p)^{-\xi} - \dfrac{\sigma}{\xi} + u, & \xi \neq 0; \\ u - \sigma \log(1-p), & \xi = 0; \end{cases}$$

Question 2

For this question you should display the R code you use to define and document ***`qgpd()`*** *within the main text of your report.*

(a) Write and document your own function `qgpd()` to calculate quantiles of a given generalised Pareto distribution. Your function should have inputs and behaviour similar to the built-in R functions such as `qnorm()` and `qunif()` and check that inputs are in the correct format.

(b) Suppose a random variable X follows an generalised Pareto distribution with threshold parameter $u = 1.5$, scale parameter $\sigma = 2$ and shape parameter $\xi = -0.4$. Use your function to find quantiles x_p for $p = 0.5, 0.75, 0.99$ (i.e. for each p find the value for which $\Pr(X < x_p) = p$).

Solution to Question 2

```
# Code for question 2

#' Computes the quantiles of a specified generalised Pareto distribution
#' @description Given vectors of probability values and vectors of
#' parameter values, returns the corresponding quantiles.
#' Recycles the values in the vectors according to the length of the
#' longest input vector as expected for built-in `R` functions and
#' set the length of the result as that.
#' @param p A numeric vector of probabilities between 0 and 1
#' @param shape A numeric value specifying the shape parameter
#' @param scale A numeric value specifying the shape parameter,
#'  greater than zero
#' @param location A numeric value specifying the location parameter
#' @param lower.tail logical; if TRUE(default),
#'  probabilities are P[X<=x], otherwise, P[X>x]
#' @param log.p logical; if TRUE, probabilities p are given
#' as log(p)
#' @return Returns a vector of the quantiles corresponding
#'  to the probability values
```

```r
#'
qgpd <- function(p, shape = 1, scale = 1, location = 1,
                 lower.tail = TRUE, log.p = FALSE ){

  if (!is.numeric(p) || !is.numeric(shape) ||
      !is.numeric(scale) || !is.numeric(location)){
    stop("Non-numeric argument to mathematical function")
  }
  # Recycle values in the input vectors according
  # to the length of the longest vector
  n        <- max(length(p), length(scale),
                  length(shape), length(location))
  p        <- rep_len(p,n)
  scale    <- rep_len(scale,n)
  shape    <- rep_len(shape,n)
  location <- rep_len(location,n)

  # process the input probability vector first accordingly

  if (log.p)       p <-exp(p)
  if (!lower.tail) p <- 1-p

  # locations where the output should be set to NaN
  valid_dex <- p>=0 & p<=1 & scale >0
  # for negative and zero scale values, set the output to NaN
  out <- rep(NaN,times = n)
  out[is.na(p)] <-NA
  out[valid_dex] <- NaN

  # the case where the shape parameter is 0 (exponential case)
  first_case <- which(valid_dex & shape == 0)
  out[first_case] <- location[first_case]-scale[first_case] *
    log(1-p[first_case])

  # the case where the shape parameter is not 0
  second_case <-which(valid_dex & shape !=0)
  out[second_case] <- (scale[second_case]/shape[second_case])*
                      ((1-p[second_case])^(-shape[second_case])-1) +
                      location[second_case]

  if (any(is.nan(out))) warning ("NaNs produced")
```

```
  return (out)
}
```

```
# Testing the function
gpd_quantiles <- data.frame(
    Probability = c(0.5,0.75,0.99),
    Quantile = (qgpd(p=c(0.5,0.75, 0.99), location = 1.5,
                     scale = 2, shape = -0.4))
)
knitr::kable(gpd_quantiles, format = "pipe",
             caption = "Probabilities and their corresponding quantiles
                        computed by the `qgpd()` function")
```

Table 1: Probabilities and their corresponding quantiles computed by the `qgpd()` function

Probability	Quantile
0.50	2.710709
0.75	3.628254
0.99	5.707553

Question 3

The file `gpd_samples.csv` contains six sets of random variates generated from different generalised Pareto distributions. The details of the generalised Pareto distributions used are summarised in `gpd_parameters.csv`. Unfortunately some of the parameter sets were recorded incorrectly.

Within a single figure, construct a series of quantile-quantile plots to identify which datasets are inconsistent with their stated distributions. You should both justify your conclusions and describe your level of confidence in your findings.

Solution to Question 3

From the six quantile-quantile plots shown in Figure 1, we can immediately see that Dataset b is the most consistent with the corresponding generalised Pareto distribution, with a high level of confidence.

Looking at the shapes of the q-q plots, we can further group datasets d and e together, as almost all of the plotted points lie under the diagonal line, resulting in a noticeable deviation along all samples, which is a strong indication the datasets d and e came from different distributions that have lighter tails than the corresponding stated distributions. The large sizes of the datasets also further confirms the conclusion.

Datasets a and c follow the distribution a bit closer than d and e at the beginning, but towards the tail end deviate. However, the dataset c only contains 20 data points, meaning it is also likely that it could be attributed to chance.

Dataset f, up to around $x = 40$, fits the distribution almost perfectly. However, towards the tail end, shows a great deviation, suggesting a heavier tail than the stated record distribution.

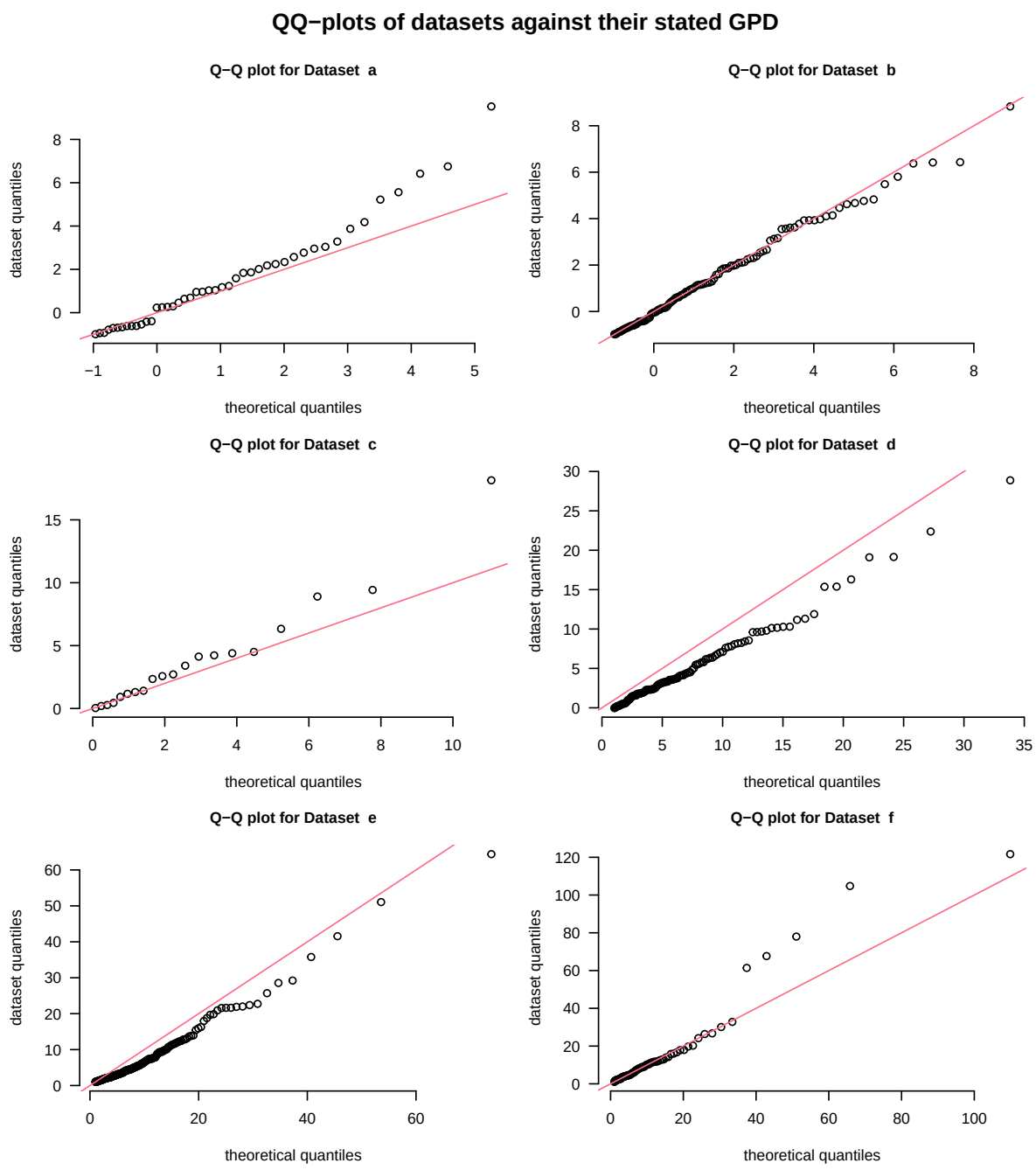


Figure 1: Quantile-quantile plots of each dataset against the quantiles of its stated generalised Pareto distribution.

Question 4

For this question, all R code should be displayed only within the appendix, not in the main report.

A hydrologist is interested in understanding the river flow at a location that is historically prone to flooding. There is a river flow gauge nearby which measures river flow in units of cubic meters per second (m^3/s or cumecs). Based on her knowledge of other rivers, the hydrologist proposes that for this river gauge:

- the distribution of river flow values is constant over time
- for river flows exceeding 75 cumecs, it is appropriate to model these data as independent and identically distributed GPD($\sigma = 29.7$, $\xi = 0.62$, $u = 0$).

Use `riverflow_2015-2024.csv` to conduct an exploratory investigation of whether these proposals are valid for the dataset provided. Summarise your findings in 250-350 words, supporting these with a collection of 4 visualisations/figures.

Solution to Question 4

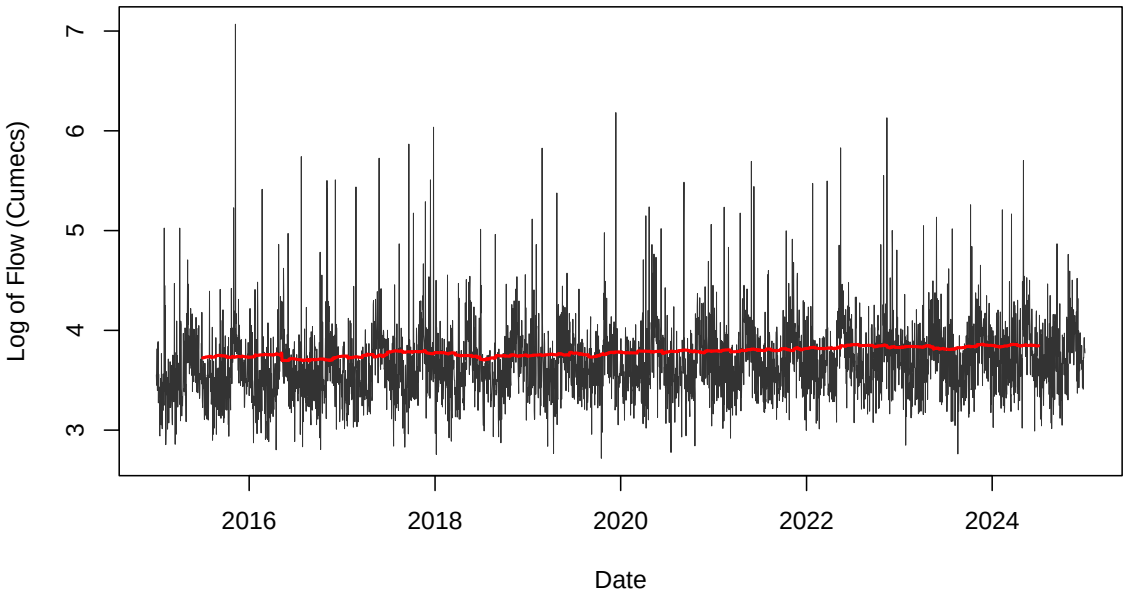
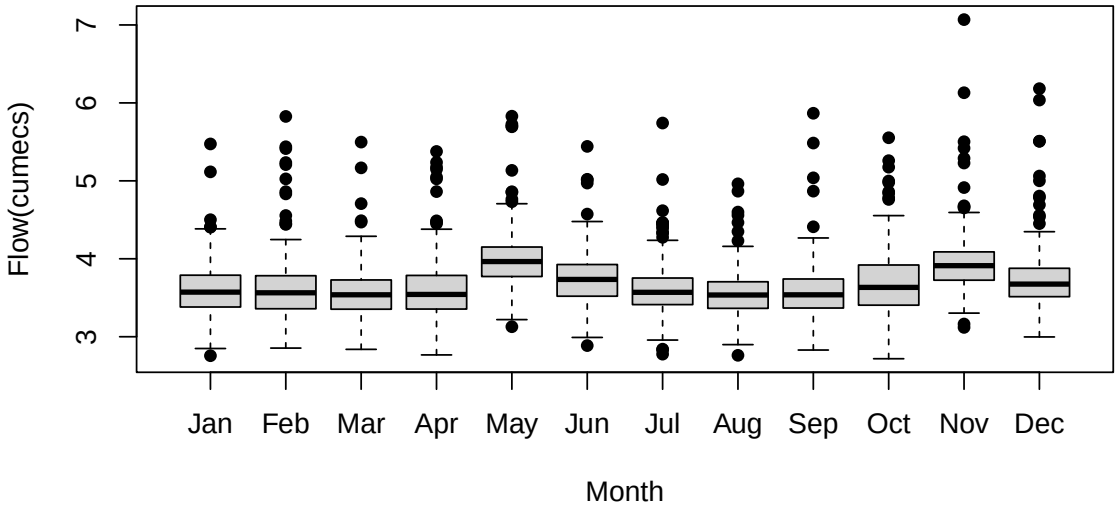
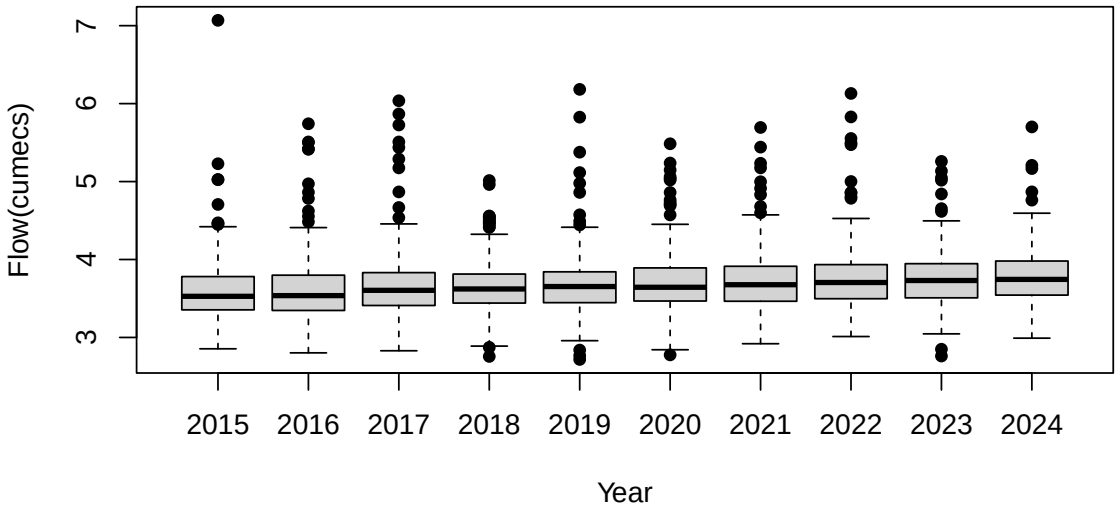


Figure 2: Time-series plot of flow over time, trend line shows rolling average



(a) By month



(b) By year

Figure 3: Box plots by month and year, log scale

We first tackle the first claim of whether the distribution of the flow is constant over time. There is evidence in the data that says otherwise. First, there is noticeable seasonality in the flow looking at Figure 3a. Namely, May and November has a significant heightening in the median flow of the river compared to other months, across the years. Furthermore, as Figure 2 shows, by the sinusoidal shape of the flow over time, that there is clear periodicity happening, likely attributed to the effects of the seasonal changes on the river.

Secondly, Looking at both Figure 2 and Figure 3b, there has been a slight but noticeable and gradual enough upward drift in the flow of the river from 2015 to 2024, evidenced by the increase in the median, lower and upper quartiles of the flows from 2015 to 2024 (Figure 3b), and also the gradual rise in the rolling average across ten years of flow data.

Therefore, there is strong evidence to suggest that the distribution of the flow of the river does not stay constant over time.

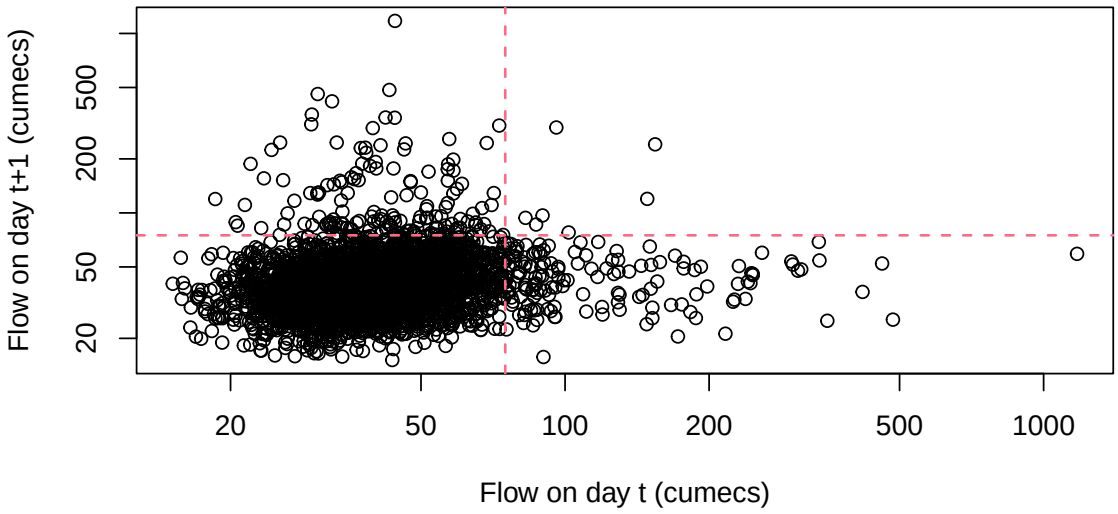


Figure 4: Lag plot showing flows between consecutive days

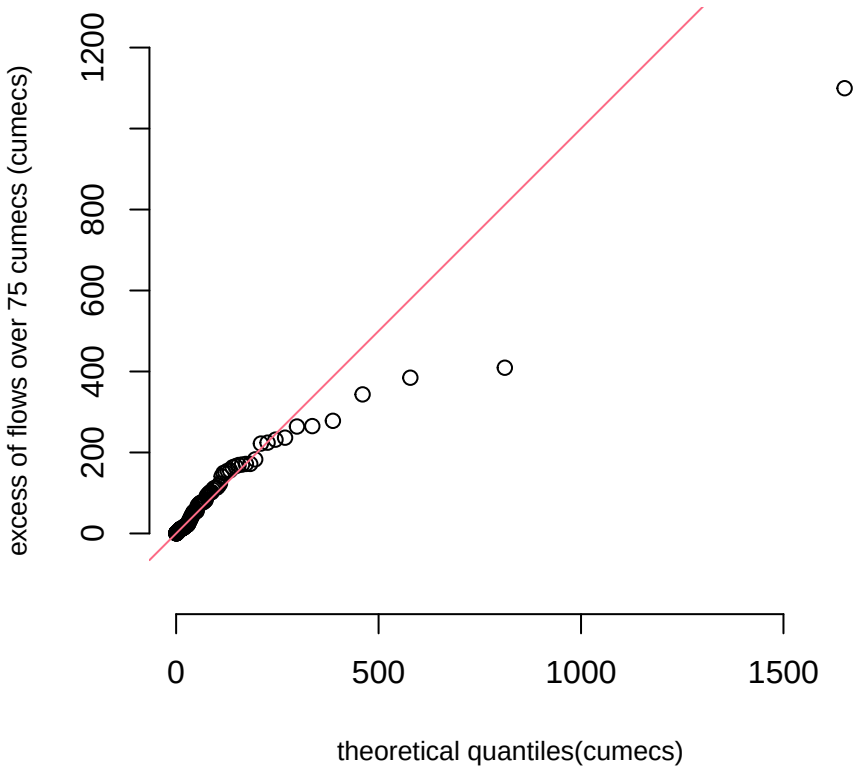


Figure 5: q-q plot for observed quantiles against the quantiles for the proposed distribution

Next, we assess the second claim of whether the excesses over 75 cumecs are following an i.i.d $GPD(\sigma = 29.7, \xi = 0.62, u = 0)$. Figure 4 shows the relationship between the flow on day t and $t + 1$. Here, the flow on day $t + 1$ is plotted against that of day t. The vertical and horizontal lines at 75 cumecs on each axis marks where the flows exceed the threshold.

As can be seen from the clustering, a great majority of points lie in the lower left quadrant, which are days of consecutive normal flows, under the threshold of 75 cumecs. It also shows that a majority days that experience heavy flows (over 75 cumecs) are isolated, as in, the river flow usually goes back to normal after having a single day of excessive flow, as can be seen from the points in the the upper left and lower right quadrants.

Finally, there are only a handful of points that lie in the upper right quadrant, which show excessive flows lasting more than a single day. Therefore, it is reasonable to assume that the excesses over 75 cumecs are independently distributed.

Figure 5 addresses whether the distribution of the observed excessive flows agrees with the suggested generalised Pareto distribution. It is then seen that the observations agree very closely with the proposed GPD for below around 450 cumecs. After that, a handful of the larger values lie below the diagonal, which indicates that the actual underlying distribution has a lighter tail than the theoretical GPD. However, this is only for a very small number of observations, so the conclusion is not definitive. It could be suggested for the hydrologist to model the excessive river flows with different parameters that would result in a better fit with the observed data.

Question 5 Reflective Summary

Describe, evidence and reflect on the process of constructing the visualisations in your answer to Question 4.

Your response should be 300-400 words and might, for example, comment on:

- why you chose these forms of visualisation and what alternatives you considered;
- a challenge you faced or something that you learned while making the visualisations;
- how you might approach this task differently with more time or for a different intended audience.

Solution to Question 5

Initially, I only had one plot that would address the second claim made by the hydrologist. In total, I had the time series plot, Figure 2, and I had counted the two box plots as two separate figures instead of two subplots in one combined figure like they are in the final version of the report. Only the qq plot, Figure 5 was included, which only answered the part of whether the underlying distribution could reasonably be modelled as a generalised Pareto distribution with the stated parameters.

However, I noticed that my plots had not addressed the claim of independence of the excesses of flows over 75 cumecs, and so I looked for ways to plot the data in such a way that would give a clear visualisation to whether independence holds. In the end, I decided to use a lag plot (Figure 4), as it is the type of plot I have the best intuition of and is also the easiest for me to read and interpret. Alternatives such as ACFs plot also came up while researching about possible ways to answer the question of independence. However, I decided to not include one as I did not have a proper understanding of the method, and so I would not have been able to provide adequate justification.

With more time, I would have looked for more ways to visualise the data that would bring more light to the underlying distribution of the river flow, the periodicity, and the nature of the excessive flows over 75 cumecs. On reflection, the time series plot also seems redundant for its intended purpose in the report when used together with the yearly box plots, Figure 3b, and the rolling mean does not in any way indicate seasonality of the flow of the river, which is visualised by Figure 3a. I would look for ways to visualise the data in a way that is not redundant. One thing that comes to my mind is to divide up the data into months and then try to show that they fit different distributions. For this more research and learning on my part should be done. All in all, stuyding this dataset and creating visualisations have taught me more about working on a dataset with a clear goal and questions to answer, and how R can be used to achieve that.

End of Assessment.

Code Appendix

```
# Code for question 2

#' Computes the quantiles of a specified generalised Pareto distribution
#' @description Given vectors of probability values and vectors of
#' parameter values, returns the corresponding quantiles.
#' Recycles the values in the vectors according to the length of the
#' longest input vector as expected for built-in `R` functions and
#' set the length of the result as that.
#' @param p A numeric vector of probabilities between 0 and 1
#' @param shape A numeric value specifying the shape parameter
#' @param scale A numeric value specifying the shape parameter,
#'  greater than zero
#' @param location A numeric value specifying the location parameter
#' @param lower.tail logical; if TRUE(default),
#'  probabilities are P[X<=x], otherwise, P[X>x]
#' @param log.p logical; if TRUE, probabilities p are given
#' as log(p)
#' @return Returns a vector of the quantiles corresponding
#'  to the probability values
#'
qgpd <- function(p, shape = 1, scale = 1, location = 1,
                 lower.tail = TRUE, log.p = FALSE ){

  if (!is.numeric(p) || !is.numeric(shape) ||
      !is.numeric(scale) || !is.numeric(location)){
    stop("Non-numeric argument to mathematical function")
  }
  # Recycle values in the input vectors according
  # to the length of the longest vector
  n        <- max(length(p), length(scale),
                  length(shape), length(location))
  p        <- rep_len(p,n)
  scale    <- rep_len(scale,n)
  shape    <- rep_len(shape,n)
  location <- rep_len(location,n)

  # process the input probability vector first accordingly

  if (log.p)       p <-exp(p)
```

```
  if (!lower.tail) p <- 1-p

  # locations where the output should be set to NaN
  valid_dex <- p>=0 & p<=1 & scale >0
  # for negative and zero scale values, set the output to NaN
  out <- rep(NaN,times = n)
  out[is.na(p)] <-NA
  out[valid_dex] <- NaN

  # the case where the shape parameter is 0 (exponential case)
  first_case <- which(valid_dex & shape == 0)
  out[first_case] <- location[first_case]-scale[first_case] *
    log(1-p[first_case])

  # the case where the shape parameter is not 0
  second_case <-which(valid_dex & shape !=0)
  out[second_case] <- (scale[second_case]/shape[second_case])*
                      ((1-p[second_case])^(-shape[second_case])-1) +
                      location[second_case]

  if (any(is.nan(out))) warning ("NaNs produced")

  return (out)
}
# Testing the function
gpd_quantiles <- data.frame(
    Probability = c(0.5,0.75,0.99),
    Quantile = (qgpd(p=c(0.5,0.75, 0.99), location = 1.5,
                     scale = 2, shape = -0.4))
)
knitr::kable(gpd_quantiles, format = "pipe",
             caption = "Probabilities and their corresponding quantiles
                        computed by the `qgpd()` function")
# Code for question 3

parameters <-read.csv ("gpd_parameters.csv")
samples <- read.csv("gpd_samples.csv")

par(mfrow = c(3, 2), mar = c(4, 4, 3, 1), oma = c(0, 0, 3, 0),
    cex.main = 0.95, las = 1)

for (cur_id in seq_len(nrow(parameters))){
```

```
    cur_parameters <- parameters[cur_id,]
    set <- cur_parameters[["id"]]
    cur_samples <- samples[samples$set_id == set,]

    qqplot(y=cur_samples$value, x=qgpd(p = ppoints(nrow(cur_samples)),
                                shape = cur_parameters[["xi"]],
                                scale = cur_parameters[["sigma"]],
                                location = cur_parameters[["u"]]),
          xlab = "theoretical quantiles",
          ylab = "dataset quantiles",
          main = paste("Q-Q plot for Dataset ", set),
          axes = FALSE
          )
    abline(0,1, col = "#fc6a85")
    axis(1)
    axis(2)
}

mtext("QQ-plots of datasets against their stated GPD", outer = TRUE,
      side = 3, line = 0.5, font = 2)

# Code for question 4
# Reading and processing the riverflow dataset
flow_data <- read.csv("riverflow_2015_2024.csv")
flow_data$date  <- as.Date(flow_data$date)

flow_data$year  <- factor(format(flow_data$date, "%Y"))
flow_data$month <- factor(format(flow_data$date, "%b"),
                          levels = month.abb)

mild_flow <- flow_data[flow_data$flow <=75,]

heavy_flow <- flow_data[flow_data$flow>75,]


flow_data$rolling <- as.numeric(stats::filter(flow_data$flow,
                                              rep(1/365,365),
                                              sides =2))

plot(
  x = flow_data$date,
  y = log(flow_data$flow),
```

```
  xlab = "Date",
  ylab = "Log of Flow (Cumecs)",
  type = "l",
  lwd = 0.4,
  col = "grey20"
)

lines(
  x = flow_data$date,
  y = log(flow_data$rolling),
  col = "red", lwd = 2
)


boxplot(log(flow_data$flow) ~ flow_data$month,
        ylab = "Flow(cumecs)",
        xlab = "Month",
        pch = 16
        )


boxplot(log(flow_data$flow) ~ flow_data$year,
        ylab = "Flow(cumecs)",
        xlab = "Year",
        pch = 16
        )

n <- length(flow_data$flow)
plot(x = flow_data$flow[-n],
     y=  flow_data$flow[-1],
     log = "xy",
     xlab = "Flow on day t (cumecs)",
     ylab = "Flow on day t+1 (cumecs)",
     pch = 1
)
abline(h = 75, v = 75, lty = 2, col = "#fc6a85", lwd = 1.5)


qqplot(x = qgpd(p = ppoints(nrow(heavy_flow)),
                shape = 0.62,
                scale = 29.7,
                location = 0),
```

```
       y = heavy_flow$flow-75,
       xlab = "theoretical quantiles(cumecs)",
       ylab = "excess of flows over 75 cumecs (cumecs)",
       cex.lab = 0.8,
       axes = FALSE,
       asp = 1
       )
abline(0,1, col = "#fc6a85")
axis(1)
axis(2)
```